Lab7

15.1 Here is the first line of a class declaration. Circle the name of the base class:

```
class Truck : public Vehicle
```

15.2 Circle the name of the derived class in the following declaration line:

```
class Truck : public Vehicle
```

Suppose a program has the following class declaration:

```
// Declaration of CheckPoint class.
class CheckPoint
{
   private:
      int a;
   protected:
      int b;
      int c;
      void setA(int x) { a = x;}
   public:
      void setB(int y) { b = y;}
      void setC(int z) { c = z;}
};
```

Answer the following questions regarding the class:

A) Suppose another class, `Quiz`, is derived from the `CheckPoint` class. Here is the first line of its declaration:

```
class Quiz : private CheckPoint
```

Indicate whether each member of the `CheckPoint` class is `private`, `protected`, `public`, or inaccessible:

a
b
c
setA
setB
setC

B) Suppose the `Quiz` class, derived from the `CheckPoint` class, is declared as

```
class Quiz : protected Checkpoint
```

Indicate whether each member of the `CheckPoint` class is `private`, `protected`, `public`, or inaccessible:

a
b
c
setA
setB
setC

C) Suppose the `Quiz` class, derived from the `CheckPoint` class, is declared as

```
class Quiz : public Checkpoint
```

Indicate whether each member of the `CheckPoint` class is `private`, `protected`, `public`, or inaccessible:

a
b
c
setA
setB
setC

11. What base class is named in the line below?

```
class Pet : public Dog
```

12. What derived class is named in the line below?

```
class Pet : public Dog
```

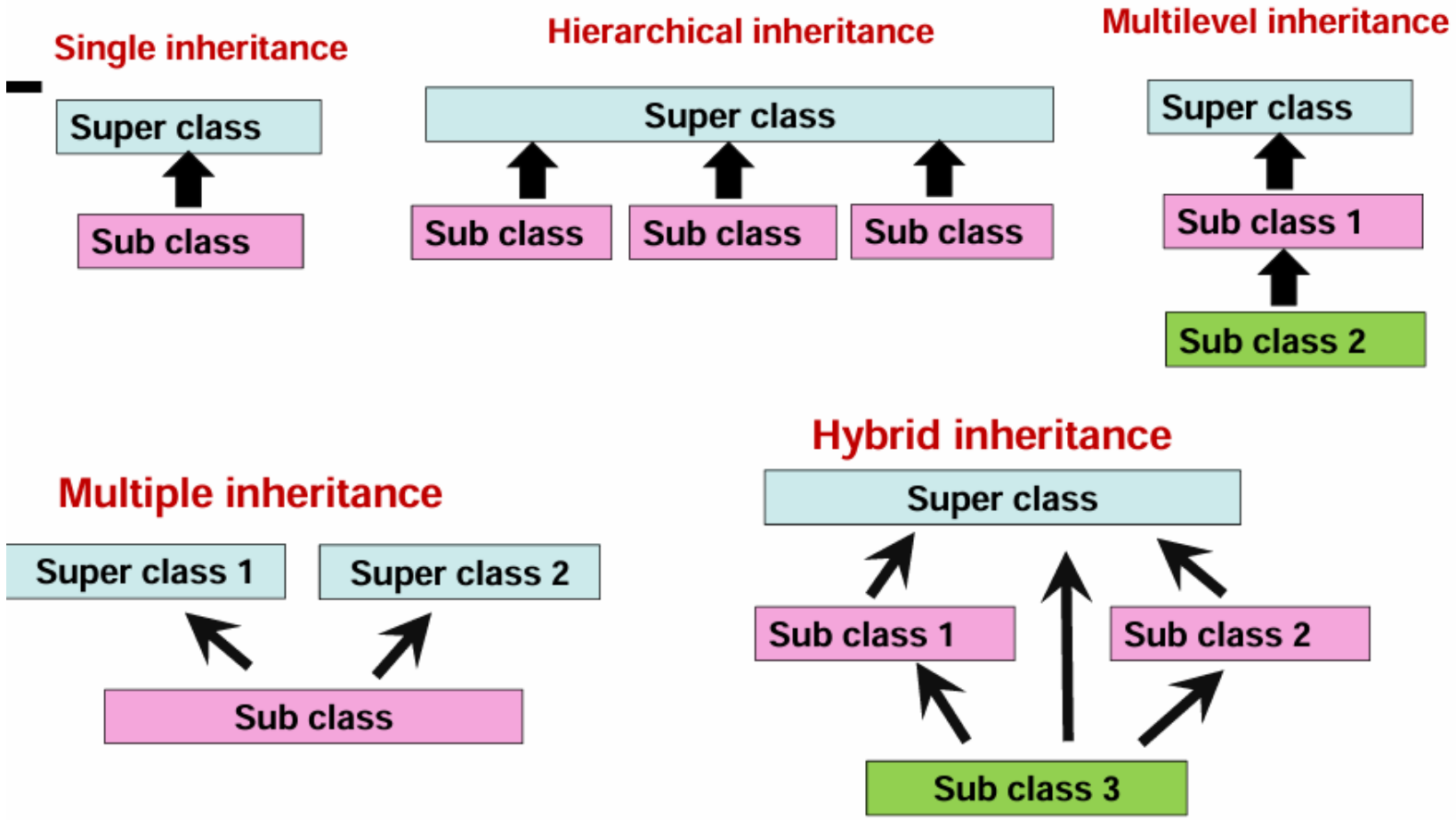
13. What is the class access specification of the base class named below?

```
class Pet : public Dog
```

14. What is the class access specification of the base class named below?

```
class Pet : Fish
```

15. Protected members of a base class are like _____ members, except they may be accessed by derived classes.


Single inheritance
Super class
Sub class
Hierarchical inheritance
Super class
Sub class
Sub class
Sub class
Multilevel inheritance
Super class
Sub class 1
Sub class 2
Multiple inheritance
Super class 1
Super class 2
Sub class
Hybrid inheritance
Super class
Sub class 1
Sub class 2
Sub class 3

Lab8

12. Ship, CruiseShip, and CargoShip Classes

Design a Ship class that has the following members:

* A member variable for the name of the ship (a string)
* A member variable for the year that the ship was built (a string)
* A constructor and appropriate accessors and mutators
* A virtual print function that displays the ship s name and the year it was built.

Design a CruiseShip class that is derived from the Ship class. The CruiseShip class should have the following members:

* A member variable for the maximum number of passengers (an int)
* A constructor and appropriate accessors and mutators
* A print function that overrides the print function in the base class. The

CruiseShip class s print function should display only the ship s name and the maximum number of passengers.

Design a CargoShip class that is derived from the Ship class. The CargoShip class should have the following members:

* A member variable for the cargo capacity in tonnage (an int).
* A constructor and appropriate accessors and mutators.
* A print function that overrides the print function in the base class. The

CargoShip class s print function should display only the ship s name and the ship s cargo capacity.

Demonstrate the classes in a program that has an array of Ship pointers. The array elements should be initialized with the addresses of dynamically allocated Ship,

CruiseShip, and CargoShip objects. (See Program 15-14, lines 17 through 22, for an example of how to do this.) The program should then step through the array, calling each object s print function.

What is (output) or (error and why)?

```
class Parent{
    private:
        int a;
    public:
        Parent(int a=0):a(a){cout <<"parent const"<< endl;}
        void SetA(int a){this->a=a;}
        int GetA(){return a;}
        virtual void Print(){cout << "from parent a = " << a << endl;}
        void X(){Print();}
        virtual ~Parent(){cout <<"parent dest"<< endl;}
};
class Child: public Parent{
    public:
        Child(int a):Parent(a){cout <<"child const"<< endl;}
        void Print(){cout << "from child a = " << GetA() << endl;}
        ~Child(){cout <<"child dest"<< endl;}
};
int main(){
    Parent* c = new Child(5);
    c->Print();
    delete c;
}
```

What is (output) or (error and why)?

```
class Parent{
    private:
        int a;
    public:
        Parent(int a=0):a(a){cout <<"parent const"<< endl;}
        void SetA(int a){this->a=a;}
        int GetA(){return a;}
        virtual void Print(){cout << "from parent a = " << a << endl;}
        void X(){Print();}
        ~Parent(){cout <<"parent dest"<< endl;}
};
class Child: public Parent{
    public:
        Child(int a):Parent(a){cout <<"child const"<< endl;}
        void Print(){cout << "from child a = " << GetA() << endl;}
        ~Child(){cout <<"child dest"<< endl;}
};
int main(){
    Parent* c = new Child(5);
    c->Print();
    delete c;
}
```

What is (output) or (error and why)?

```
class Parent{
    private:
        int a;
    public:
        Parent(int a=0):a(a){cout <<"parent const"<< endl;}
        void SetA(int a){this->a=a;}
        int GetA(){return a;}
        void Print(){cout << "from parent a = " << a << endl;}
        void X(){Print();}
        ~Parent(){cout <<"parent dest"<< endl;}
};
class Child: public Parent{
    public:
        Child(int a):Parent(a){cout <<"child const"<< endl;}
        void Print(){cout << "from child a = " << GetA() << endl;}
        ~Child(){cout <<"child dest"<< endl;}
};
int main(){
    Parent* c = new Child(5);
    c->Print();
    delete c;
}
```

```
class Parent{
    private:
        int a;
    public:
        Parent(int a=0):a(a){cout <<"parent const"<< endl;}
        void SetA(int a){this->a=a;}
        int GetA(){return a;}
        void Print(){cout << "from parent a = " << a << endl;}
        void X(){Print();}
        ~Parent(){cout <<"parent dest"<< endl;}
};
class Child: public Parent{
    public:
        string *prt;
        Child(int a):Parent(a){cout <<"child const"<< endl; *ptr = new int(5);}
        void Print(){cout << "from child a = " << GetA() << endl;}
        ~Child(){cout <<"child dest"<< endl; delete ptr;}
};
int main(){
    Parent* c = new Child(5);
    c->X();
    delete c;
}
```

1- After execution, why the ptr pointer in Child is not deleted?

2- Fix the program to make “delete ptr pointer” executed

What is (output) or (error and why)?

```
class Parent1{
    public:
        void Print(){cout << "from parent1" << endl;}
        ~Parent1(){cout <<"parent1 dest"<< endl;}
};
class Parent2{
    public:
        void Print(){cout << "from parent2" << endl;}
        ~Parent2(){cout <<"parent2 dest"<< endl;}
};
class Child: public Parent1, public Parent2{
    public:
        ~Child(){cout <<"child dest"<< endl;}
};
int main(){
    Child c;
    c.Print();
}
```

What is (output) or (error and why)?

```
class Parent1{
    public:
        void Print(){cout << "from parent1" << endl;}
        ~Parent1(){cout <<"parent1 dest"<< endl;}
};
class Parent2{
    public:
        void Print(){cout << "from parent2" << endl;}
        ~Parent2(){cout <<"parent2 dest"<< endl;}
};
class Child: public Parent1, public Parent2{
    public:
        ~Child(){cout <<"child dest"<< endl;}
};
int main(){
    Parent1* c = new Child();
    c->Print();
    delete c;
}
```